

Walid Rashad – Candidate for the Engineering Profession (OIQ)

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[Linkedin](#)

WORK PROFILE

Electrical Engineering graduate (Magna Cum Laude, University of Ottawa) and Candidate for the Engineering Profession (CEP) within the OIQ in Quebec (Eligible for registration as a Professional Engineer). Over one year of work experience with specialization in embedded systems, hardware design and PCB design. Acquired knowledge on CAN BUS networks and electrical system integration.

I have experience in MATLAB/Simulink, in which I performed simulations of control, power, and signals/systems. I have an excellent capacity for innovation and problem solving. I can work under pressure with tight deadlines and high demands, being able to prioritize my tasks. I have worked on several multidisciplinary projects with engineering teams, through the stages of design, cost/risk estimation, prototyping, validation, production and evaluation.

SKILL SUMMARY

- Over 1 year of professional experience in engineering – CEP certification acquired (OIQ).
- **Engineering skills:** FPGAs, Embedded Systems, Hardware Engineering, Systems Engineering, microcontrollers, PCB design, Analog and Digital Circuits, Design Documentation, Lab Equipment (oscilloscopes, multimeters, soldering, filters, logic analyzers, etc.), PCB Design, Data Acquisition Systems, Automation/Robotics (kinematics and inverse kinematics)
- **Programming:** Raspberry Pi/Linux, Arduino, Github, Python, C/C++, Assembly (PIC16F917), HTML, CSS, JavaScript, MATLAB/Simulink, VBA.
- **Software:** MS Visual Studio Code, MATLAB/Simulink, Altium/KiCad, Multisim, Microsoft Office 365.
- **Communication:** Bilingual (French and English) and fluent. Interpersonal.
- **Valid driving license – can work on site/relocate. Willing to travel abroad.**
- **Acquired Knowledge:** CAN BUS networks/aircraft electrical systems

EDUCATION

University of Ottawa, Ottawa (Ontario) - Bachelor's degree in Electrical Engineering - Magna Cum Laude

September 2020 – April 2025

WORK EXPERIENCE

University of Ottawa – Teaching Assistant

January 2025 – April 2025

- Programmed and debugged 'PICDEM' mechatronics boards using PIC16F917 microcontrollers.
- Employed lab equipment such as oscilloscopes to measure motor output voltage
- Supervised and executed lab sessions for students.
- Marked lab work and exams.
- Mentored students in reasonable and engaging manners.

Innovation, Science and Economic Development - Telecommunications Officer

September 2023 – December 2023

- Programmed VBA macros to improve the efficiency of Excel/Access databases.
- Approved deliverables submitted by high-profile clients for milestones to facilitate subsidies
- Approved of network infrastructure projects and coverage maps.
- Trained personnel on geospatial mapping software.
- Cleared for Secret Security clearance for 10 years.

Bombardier Inc. - Electromagnetic Compatibility Intern

January 2023 – April 2023

- Performed Electromagnetic interference simulations on aircraft harnesses.
- Designed, validated tools for electromagnetic compatibility (EMC) analysis in harness segregation projects and created tutorials for future reference.
- Developed Python scripts to analyze 20+ EMC test cases, identifying trends and anomalies in interference behavior.

- Reviewed and optimized 50+ aircraft electrical schematics to improve embedded system reliability and EMC compliance.
- Monitored Aircraft structure integrity and ensured compliance with airworthiness requirements.

Canadian Coast Guard - Co-op Student in Electrical Engineering

May 2022 – August 2022

- Identified and corrected 60+ electrical system risks and non-compliance issues across marine vessels.
- Drafted and validated electrical schematics for shipboard systems.
- Researched renewable energy solutions for marine applications.
- Developed risk-based strategies for integrating renewable energy into marine platforms.

ENGINEERING PROJECTS

- **FSAE Formula Club**
 - Learning and optimizing printed circuit boards (PCBs) with Altium.
 - Managing the lambda ratio (fuel map) in Race Studio 3
 - Soldering and integration of electrical systems
- **Designed a Raspberry Pi Data Acquisition System to translate Morse code into text.**
 - Used a geophone to detect vibrations and transform them into binary data
 - Used a low-pass filter to remove noise and unwanted vibrations
 - Converted voice to text using Python
- **Developed a search and rescue robot with Raspberry Pi, equipped with multiple sensors.**
 - Developed Rocker-Bogie suspension system
 - Developed the Interface and programmed the thermal camera, LIDAR sensor and motors
 - Developed the circuitry (motors, camera, battery, buck converter, etc.)
- **Created a portable personal finance tracker using Arduino Nano.**
 - Designed PCB from scratch
 - Safe fireproof wiring and soldering through PCB
 - 3D modeling of case
 - Connected screen, keypad, battery, SPDT switch and buck converter in case
- **Modeled PID Controllers in Simulink.**
 - Water level sensor
 - Temperature control
 - Platform stabilization